

Mental Health in Tech Analytics

Executive Analytics Report

2014 Mental Health in Tech Survey • 1,259 Respondents • EDA + Statistical Validation + Streamlit Dashboard

1,259

Respondents Analyzed

5

Validated Factors

6

Dashboard Modules

1

Live App Shipped

1. Executive Summary

This project analyzes the 2014 Mental Health in Tech Survey to understand which demographic, personal, and workplace factors are associated with mental health treatment-seeking behaviour among technology employees. The analysis combines data cleaning, feature engineering, exploratory data analysis, multivariate analysis, correlation analysis, and statistical validation using Chi-square tests and Cramer's V.

TOP SIGNAL

Work Interference — 85.4% vs 14.1% treatment rate

The strongest analytical signal in the entire dataset. Respondents who report mental health often interferes with work are treated at nearly 6x the rate of those who report it never does.

Family history was the second strongest treatment-related factor, with treatment rates of 74.2% for respondents reporting a family history versus 35.5% for those without one. The analysis also identifies meaningful relationships involving mental health care options, employer benefits, and gender. Statistical testing was used to distinguish stronger relationships from variables that did not demonstrate statistically significant associations. The resulting insights were translated into an interactive, multi-page Streamlit dashboard for stakeholder-oriented exploration.

2. Project Objective

The primary objective is to identify and validate the factors most strongly associated with mental health treatment behaviour in the technology workplace and translate those findings into practical workplace insights.

- Understand patterns in mental health treatment-seeking behaviour.
- Identify groups with substantially different treatment rates.
- Evaluate workplace support mechanisms such as benefits and care options.
- Examine how work interference relates to treatment behaviour.
- Assess workplace culture and attitudes surrounding mental health.
- Use statistical validation rather than relying only on visual differences.
- Translate analytical findings into practical recommendations for workplace support.

3. Dataset & Analytical Scope

The project uses the 2014 Mental Health in Tech Survey, an Open Sourcing Mental Illness dataset designed to examine attitudes toward mental health and the frequency of mental health disorders in the technology workplace. The source contains demographic, employment, mental health, workplace-support, and workplace-attitude variables.

- Dataset size: 1,259 respondents and 27 original variables.
- Target/outcome variable: treatment — whether the respondent reported seeking treatment for a mental health condition.
- Key dimensions: age, gender, country/state, family history, work interference, company size, remote work, benefits, care options, anonymity, leave, workplace culture, and mental-health perceptions.
- The source dataset itself identifies geographic variation and workplace predictors/attitudes as key exploratory questions.

4. Data Preparation & Feature Engineering

- Invalid age values were converted to missing, while valid ages were retained within the observed 18-72 range.
- Free-text gender responses were standardized into Male, Female, and Other.
- Missing state and self-employment values were retained as Unknown rather than artificially imputed.
- Missing work_interference responses were labelled Not Answered and kept distinct from Never.
- The comments field was excluded from quantitative analysis because of very high missingness.
- Age Group, Company Size Group, Remote Work Status, Mental Health History, Treatment Status, and Work Interference Level were engineered for analysis.

Data Preparation

- ☒ Invalid Values Handled — age outliers converted to missing without dropping respondents
- ☒ Categories Standardized — free-text gender responses cleaned and unified
- ☒ Missingness Respected — no artificial imputation on Unknown or Not Answered
- ☒ Feature Engineer — 6 new analytical variables created

5. Analytical Approach

5.1 Univariate Analysis

The analysis first established the distribution of respondent demographics, treatment status, family history, work interference, mental health benefits, care options, perceived anonymity, and geographic representation.

5.2 Bivariate Analysis

Treatment rates were compared across major demographic, personal, and workplace factors. This stage identified large treatment-rate differences and highlighted potential drivers for further statistical validation.

5.3 Multivariate Analysis

Multivariate views examined combinations of work interference and family history, as well as mental-health benefits and care options. A Spearman correlation matrix was also used for appropriately encoded ordinal variables.

5.4 Statistical Validation

Chi-square tests of independence and Cramer's V were used to assess whether observed differences represented statistically significant associations and to quantify effect strength. Work interference was validated without treating Not Answered as a substantive interference category.

6. Key Findings

Ranked by effect size (Cramer's V). Larger values indicate a stronger, more statistically defensible association with treatment-seeking behaviour.

Rank	Factor	Key Result	Cramer's V	Strength
1	Work Interference	85.4% Often vs 14.1% Never	0.542	STRONG
2	Family History	74.2% with vs 35.5% without	0.375	STRONG
3	Care Options	Treatment differs by awareness	0.272	MODERATE
4	Mental Health Benefits	Treatment differs by benefit access	0.223	MODERATE
5	Gender	Statistically significant differences	0.204	MODERATE

The statistical results indicate that work interference has the strongest association with treatment among the tested factors, followed by family history, care options, benefits, and gender.

7. Work Interference: Strongest Treatment-Related Signal

Treatment rates rise sharply as reported work interference increases. The observed rates are approximately 14.1% for Never, 70.5% for Rarely, 77.0% for Sometimes, and 85.4% for Often.

The difference between Often and Never is approximately 71 percentage points. The statistical analysis supports a strong association, with Cramer's V of approximately 0.542.

BUSINESS INTERPRETATION

Treat interference as an early-support signal

Work interference can be treated as an important signal for strengthening early support and accessible care pathways. This should not be interpreted as evidence that work interference causes treatment-seeking.

8. Family History: Second Strongest Association

Respondents reporting a family history of mental health conditions had a treatment rate of approximately 74.2%, compared with 35.5% among respondents without a reported family history. The corresponding Cramer's V of approximately 0.375 indicates a substantial association.

BUSINESS INTERPRETATION

Population-level signal, not an individual label

Organizations should avoid using family history for individual profiling. At a population level, however, the result reinforces the value of accessible, stigma-aware mental health resources.

9. Workplace Support & Culture

The project evaluates mental health benefits, care options, wellness programs, resources to seek help, anonymity, medical leave, coworker and supervisor attitudes, perceived consequences of disclosure, interview attitudes, and perceptions of mental versus physical health.

- Awareness of care options is significantly associated with treatment behaviour.
- Reported access to mental health benefits is significantly associated with treatment behaviour.
- Workplace culture variables provide additional context around disclosure, stigma, and perceived consequences.
- The dashboard allows stakeholders to explore these dimensions interactively rather than relying on a single static view.

10. Geographic Analysis

The project includes worldwide respondent distribution and a United States state-level view. The analysis establishes where the survey sample is concentrated and provides geographic context for interpreting the findings.

However, the geographic question from the original dataset is only partially addressed. The project does not claim a comprehensive country/state-by-attitude analysis because the sample is strongly concentrated in the United States. Geographic results should therefore be interpreted as sample representation rather than global prevalence.

11. Non-Significant Findings

Several candidate factors did not show statistically significant associations with treatment in the final validation:

No Significant Association Found

- ☐ Age Group
- ☐ Remote Work Status
- ☐ Company Size
- ☐ Supervisor Support
- ☐ Physical Health Interview
- ☐ Physical Health Consequence

These results are important because they prevent the analysis from over-interpreting every visible difference. They support a more evidence-based approach in which interventions are prioritized around stronger signals rather than assumptions.

12. Dashboard & Deployment

The analytical work has been converted into a deployed multi-page Streamlit dashboard. The application provides interactive exploration across respondent demographics, workplace support and culture, mental health and treatment, statistical analysis, and executive insights.

-  Overview — project context, KPIs, and high-level findings.
-  Respondent Profile — demographic and geographic analysis.
-  Workplace Support & Culture — benefits, care options, wellness programs, anonymity, leave, and workplace attitudes.
-  Mental Health & Treatment — treatment patterns, family history, work interference, and treatment gaps.
-  Statistical Analysis — statistical significance and Cramer's V.
-  Executive Insights — concise findings and business recommendations.

 Live Dashboard: <https://mental-health-in-tech-analysis-kartikey-singh.streamlit.app/>

13. Business Recommendations

Prioritize early support

Use work-related interference as a population-level signal for improving early support and referral pathways.

Improve care visibility

Communicate available mental health care options clearly so employees understand where and how to seek support.

Strengthen benefit communication

Ensure employees understand what mental health benefits cover and how to access them.

Reinforce confidentiality

Make privacy and anonymity policies easy to understand and visible to employees.

Build supportive culture

Reduce stigma around disclosure and improve the workplace environment surrounding mental health discussions.

Avoid demographic profiling

Do not use age, family history, gender, or other characteristics to label individuals; use analytics to improve population-level support systems.

14. Limitations & Responsible Interpretation

- The dataset was collected in 2014 and may not represent current workplace conditions.
- The survey is self-reported and observational.
- The sample has strong U.S. representation and is not globally representative.
- Treatment rate is an outcome measure and should not be interpreted as mental-health prevalence.
- Statistical relationships represent associations, not causal effects.
- The comments field contains substantial missing data and was excluded from quantitative analysis.
- Small geographic groups should not be used for strong comparative conclusions.

15. Portfolio Value

From a Data Analyst perspective, the project demonstrates an end-to-end workflow: transforming raw survey data, cleaning and standardizing variables, engineering analytical features, exploring patterns visually, validating relationships statistically, communicating findings, and converting the analysis into an interactive deployed product.

 Python-based data analysis	 Data cleaning & preprocessing
 Feature engineering	 Exploratory & multivariate analysis
 Statistical hypothesis testing	 Effect-size interpretation
 Interactive visualization	 Streamlit dashboard development
 Business insight generation	 Data storytelling & executive communication

16. Conclusion

The Mental Health in Tech Analytics project provides a structured, evidence-based view of treatment-seeking behaviour in the 2014 technology workplace survey. The strongest signal is work interference, followed by family history, while workplace support mechanisms such as care options and mental health benefits also show meaningful associations.

The project goes beyond descriptive charts by applying statistical validation and explicitly distinguishing significant from non-significant relationships. It also translates the analysis into a deployed Streamlit dashboard, making the results accessible through interactive exploration.

The central business takeaway is that organizations can create stronger mental health support systems by focusing on accessibility, awareness, confidentiality, and early support — while avoiding individual-level profiling and recognizing that the survey findings represent associations rather than causal effects.

 STATUS

End-to-end analysis delivered: cleaned, validated, visualized, and deployed.

17. Project Links

 Live Streamlit Dashboard: [Mental Health in Tech Analytics](#)

 GitHub Repository: [Mental Health in Tech Analytics](#)